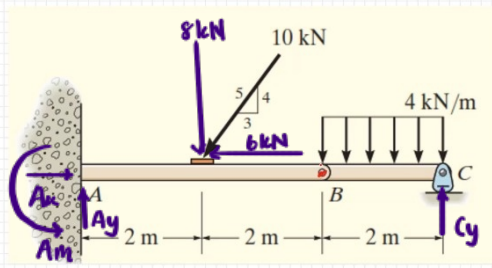


## Example 1:

A 'frame' comprises two horizontal members connected by a pin at B.

Determine the reaction forces at A and C and the internal pin forces at B. Go on to draw the SFD and BMD for the frame system.



### Entire System

$$\sum F_x = A_x - 6$$

$$A_x = 6 \text{ kN} \quad \#$$

$$\sum F_y = A_y + C_y - 8 - 4(2)$$

$$A_y + C_y = 18 \quad \text{--- (1)}$$

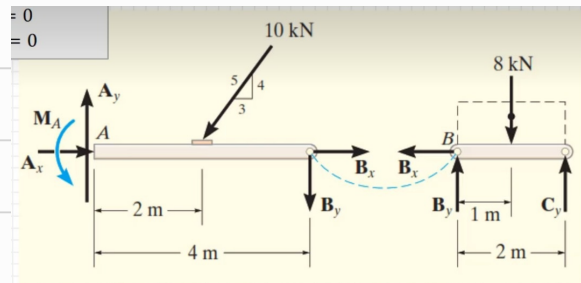
$$\sum M_A = A_m - (8 \times 2) - 8(5) + 6C_y$$

$$A_m + 6C_y = 56 \quad \text{--- (2)}$$

Since we have 4 unknowns [ $A_m, A_y, C_y, A_x$ ]

but only 3 eqns, this FBD of the entire system is not enough.

### Each Member



Since RHS member have less unknown, we look at it 1<sup>st</sup>.

$$\sum F_x = B_x \implies B_x = 0$$

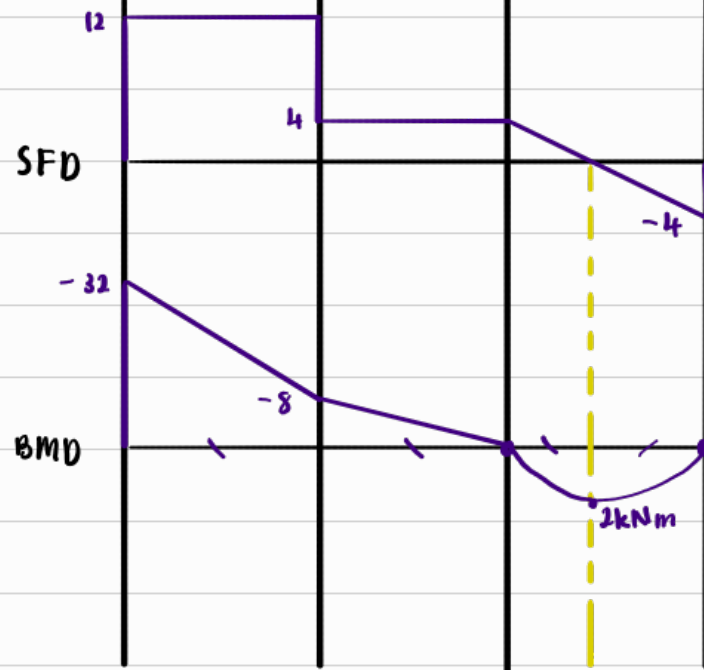
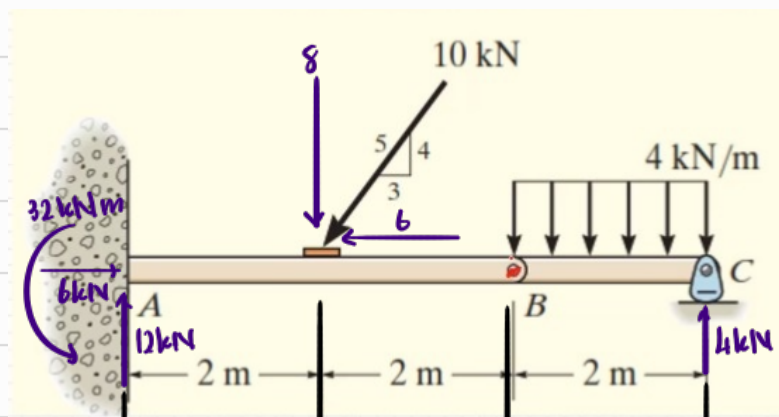
Since the system is symmetrical,

$$B_y = C_y = \frac{8}{2} = 4 \text{ kN} \quad \text{--- (3)}$$

$$\text{(3)} \rightarrow \text{(1)} \text{ \& } \text{(2)}$$

$$A_y = 12 \text{ kN}$$

$$A_m = 32 \text{ kN}$$

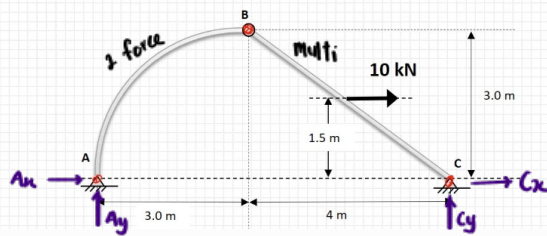


$$\frac{dM}{dx} = V$$

## Example 2:

A 3-pin arched 'frame' comprises two members one straight and one curved connected by a pin at B. There are simple pin supports at A and C. A horizontal load is applied as shown. Assume member AB follows the shape of a circular arc. Neglect member self-weight here

- Confirm which members are multi-force and which are two force members?
- Calculate the support reaction forces and the internal pin forces at B
- Calculate the maximum bending moment in each member (if there is any)?



### b) Entire diagram

$$\sum M_A = -10(1.5) + C_y(3+4)$$

$$C_y = \frac{15}{7} \approx 2.142 \text{ kN}$$

$$\sum F_y = A_y + C_y$$

$$A_y = -C_y = -\frac{15}{7} \text{ kN}$$

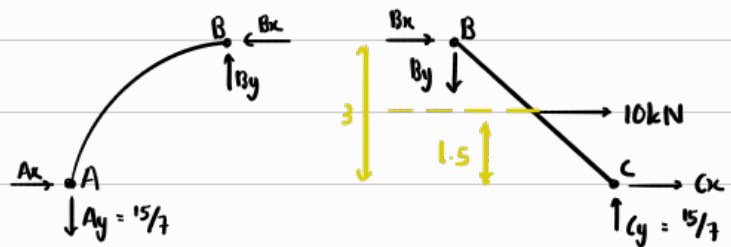
$$\sum F_x = A_x + 10 + C_x$$

$$A_x + C_x + 10 = 0 \quad \text{--- ①}$$

Since we have 4 unknowns  $[A_x, C_x, A_y, C_y]$

but only 3 eqns, this FBD of the entire system is not enough.

### Each member



Looking at RHS member,

$$\sum M_B = 10(1.5) + \frac{15}{7}(4) + 3C_x$$

$$C_x = -7.857 \text{ kN} \quad \text{--- ②}$$

② → ①

$$A_x = -2.142 \text{ kN} = -\frac{15}{7} \text{ kN}$$

Looking at LHS member,

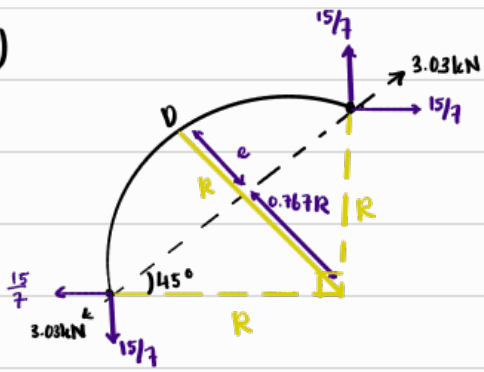
$$\sum F_x = A_x - B_x$$

$$B_x = -\frac{15}{7} \text{ kN}$$

$$\sum F_y = B_y - \frac{15}{7}$$

$$B_y = \frac{15}{7} \text{ kN}$$

c)



$$e = R - 0.767R = 0.233R = 0.233(3) = 0.700\text{ m}$$



$$\text{Maximum moment, } M = 3.03(e) = 3.03(0.700) = 2.121\text{ kNm}$$

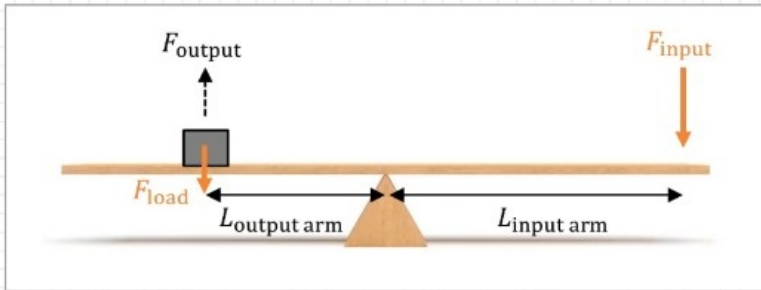


# Mechanical advantage

A lever in equilibrium ( $\sum M = 0$ ) balances the input and the load

$$F_{\text{input}} L_{\text{input arm}} = F_{\text{load}} L_{\text{load arm}}$$

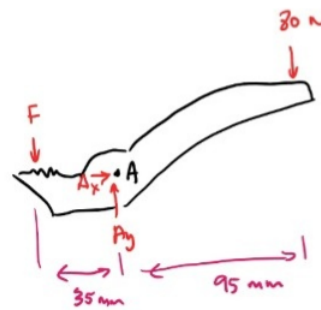
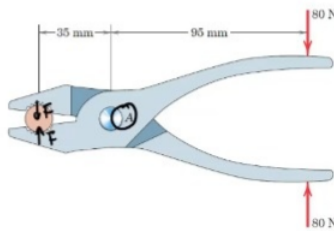
Mechanical advantage ( $MA$ ) is a measure of the ratio of output force to input force



$$MA = \frac{F_{\text{output}}}{F_{\text{input}}} = \frac{F_{\text{load}}}{F_{\text{input}}} = \frac{L_{\text{input arm}}}{L_{\text{load arm}}}$$

## Example

applied on the rod by each of the plier jaws. Additionally, calculate the forces in the pin at point A. Finally, determine the mechanical advantage of this simple machine.



$$\sum F_x = 0$$

$$A_x = 0$$

$$\sum F_y = 0$$

$$-F + A_y - 80 = 0$$

$$-F + A_y = 80$$

$$A_y = 80 + F$$

$$= 80 + 217.1429$$

$$A_y = 297.1429 \text{ N}$$

$$\sum M_A = 0$$

$$F(35) - 80(95) = 0$$

$$F = 217.1429 \text{ N}$$

$$MA = \frac{F_{\text{output}}}{F_{\text{input}}} = \frac{217.1429}{80}$$

$$MA = 2.7143$$